

DIRECT TRANSIENT STABILITY ASSESSMENT WITH EXCITATION CONTROL

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Abstract: Recently the authors incorporated the effect of excitation control on the transient energy of a multi-machine power system during the first swing. The models used are: the generator represented by the two-axis model and the exciter represented by one gain, one time constant and one limiter.

In this paper the parameters of the newly developed transient energy function, which assumes a constant average internal EMF, are computed. The critical value of this transient energy is determined and its use in transient stability assessment is investigated.

A procedure for transient stability assessment using the new energy function has been developed and successfully applied to fault disturbances in two test networks.

I. INTRODUCTION

In the literature on direct stability analysis, it is known that accurate direct transient stability assessment requires a transient energy function which identifies the transient energy tending to separate some generators from the rest, and a good estimate of its critical value. Recently, the authors have extended the application of the transient energy function (TEF) method by incorporating the exciter effects in the system energy in the first swing transient [1]. In this paper, the critical value of the new transient energy function is determined and is used for transient stability assessment. There have been some approaches, previously discussed in the literature, in which "a higher order" synchronous generator model is incorporated in direct transient stability analysis methods [2-7]. These approaches, are based on system-theoretic arguments. The emphasis is analytical in nature. Application to practical power system problems has not been reported in the literature.

In Section 6 of reference [2], the problem of detailed modeling has been approached with a more practical point of view. The investigation is primarily concerned with the study and analysis of the system transient energies and the effects of the internal voltage variations on these energies for a specific multi-machine test system. The generator model used was the two-axis representation. Transient saliency and saturation were neglected. Excitation system effects were included in the model. This entire study consisted mainly of analyzing the energy patterns. It was found that the invariance of the potential energy at the boundary surface crossing does not hold due to the change of generator flux.

II. GENERAL APPROACH

The fundamental approach followed in developing a model for incorporating the exciter effect in the first swing transient is that the transient is an inertial transient. The generators' motion is determined by the combination of (or interaction among) the synchronizing torques and the generators' inertias. The exciters primarily alter the generators' synchronizing torques.

Main Assumptions

There are some basic assumptions used with the above approach. They are [1]:

1. Generator flux:
 - i. Does not change abruptly in the first swing, i.e., a reasonably well-behaved function can describe the flux variation, on the average, in the first swing transient.
 - ii. If the conditions initially (e.g., at the end of the disturbance) and at the peak of the swing are known, only a "smooth" variation of the flux between them takes place.
2. The key parameters needed to compute the transient energy can be obtained from two solutions: one at clearing and the other at the point where the system potential energy is maximum on the critically stable trajectory. In the rest of the paper this point will be referred to as the peak point.
3. When the disturbance is removed the total transient energy is constant.

III. THE POWER SYSTEM MODEL

Synchronous Generator Model

A synchronous generator with exciter is assumed to be represented by the so-called two-axis model (see Section 4.15.3 of [8]). Figure 1 shows a schematic of the model:

x' is the transient reactance (sometimes an average of x'_d and x'_q), E' is the voltage behind

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x' and is not constant; its components E'_d and E'_q vary and have different time constants

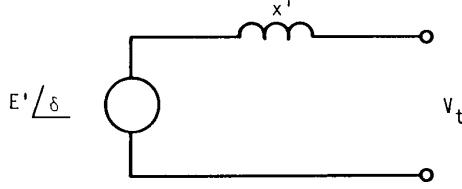


Figure 1. Synchronous Generator Model

The remaining synchronous generators are to be modeled classically, i.e., with E' (in Figure 1) constant.

Excitation System Model

The excitation system is assumed to be represented by an equivalent one gain, one time constant, and one limiter model.* It is to be noted that there are standard techniques available in the literature (see for example Section 3.5 of [9]) to reduce a higher order system to an equivalent one gain, one time constant system in the frequency range of interest. The exciter model used is shown in Figure 2. We also note that by proper choice of the parameters K_A , τ_E , E_{FDmax} , and E_{FDmin} most modern excitation systems, and hence, the desired flux variation can be modeled.

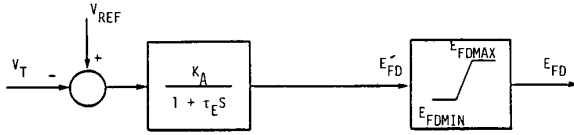


Figure 2. Exciter Model

The Power Network

Except for the generators represented in detail, the rest of the power network is represented classically, i.e., with loads represented by constant impedance.

It is also assumed that the network is reduced to the internal generator nodes.

Generator-Exciter Equations

With the generator modeled by the two-axis model, the exciter represented by one time constant, the generator-exciter equations for generator k in the COI frame are given by (see Chapter 8 of reference [8]).

$$\tau'_{dok} \dot{E}'_{qk} = E_{FDk} - E'_{qk} + (x_{dk} - x'_{dk}) I_{dk} \quad (1)$$

$$\tau'_{qok} \dot{E}'_{dk} = -E'_{dk} - (x_{qk} - x'_{qk}) I_{qk} \quad (2)$$

$$\tau_{Ek} \dot{E}'_{FDk} = g(E'_{FDk}, V_{tk}, V_{REFk}) \quad (3)$$

$$\begin{aligned} &= -E'_{FDk} + K_A (V_{REFk} - V_{tk}) \\ \dot{M}_k \ddot{\omega}_k &= P_{Mk} - E'_{dk} I_{dk} - E'_{qk} I_{qk} + (x'_{qk} - x'_{dk}) I_{dk} I_{qk} - \frac{M_k}{M_T} P_{COI} \end{aligned} \quad (4)$$

$k=1, 2, \dots, m.$

where

m = No. of generators represented in detail.

τ'_{do}, τ'_{qo} = Open circuit direct and quadrature axes time constants respectively.

E_{FD}, E'_{FD} = Stator emf corresponding to field voltage. E_{FD} =value after the limiter; E'_{FD} =value before the limiter.

x_d, x'_{dk} = Direct axis synchronous and transient reactances.

x_q, x'_{qk} = Quadrature axis synchronous and transient reactances.

E'_d, E'_q = Direct and quadrature axes stator emfs corresponding to rotor flux components.

I_d, I_q = Direct and quadrature axes stator currents.

V_t = Generator terminal voltage.

V_{REF} = Exciter reference voltage.

We also note that if there are $(n-m)$ generators represented classically, the equations describing them would be

$$M_k \ddot{\omega}_k = P_{Mk} - E_k I_{qk} - \frac{M_k}{M_T} P_{COI} \quad (5)$$

$k = m+1, \dots, n$

where E = voltage behind transient reactance,

n = total no. of generators in the system.

In addition, the equations relating the currents and voltages complete the description of the system (see Chapter 9 of [8]).

IV. TRANSIENT ENERGY FUNCTION FORMULATION

Following the detailed derivation in [1], based on the following assumptions:

1. A linear trajectory approximation is used to evaluate the path dependent terms, and
2. the internal generator voltages E'_d and E'_q , are treated as constants:

- a) for a generator with exciter, an average between E' at clearing and at the peak point, and
- b) for a generator represented classically $E'_{qi} = E_i$, and $E'_{di} = 0$;

Accounting only for the energy corresponding to the synchronizing torques, the potential energy is given by

$$\begin{aligned} V_{PE} &= \sum_{i=1}^n P_{mi} (\theta_i - \theta_i^s) + \sum_{i=1}^n \alpha_{ii} G_{ii} (\theta_i - \theta_i^s) + \sum_{i=1}^{n-1} \sum_{j=i+1}^n B_{ij} \\ &\quad [\alpha_{ij} (-\cos \theta_{ij} + \cos \theta_{ij}^s) + \beta_{ij} (\sin \theta_{ij} - \sin \theta_{ij}^s)] \\ &\quad + \sum_{i=1}^{n-1} \sum_{j=i+1}^n G_{ij} \frac{\theta_i + \theta_j - \theta_i^s - \theta_j^s}{\theta_{ij} - \theta_{ij}^s} [\alpha_{ij} (\sin \theta_{ij} - \sin \theta_{ij}^s) \\ &\quad + \beta_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^s)] \end{aligned} \quad (6)$$

* This model is used for convenience and to standardize the treatment of incorporation of the effect of flux variation in TEF. A higher order model can be used, if desired.

where $\alpha_{ij} = E'_{di} E'_{dj} + E'_{qi} E'_{qj}$,

$$\beta_{ij} = E'_{di} E'_{qj} - E'_{qi} E'_{dj} \quad (7)$$

For a generator k represented by the classical model, $E'_{dk} = 0$, and $E'_{qk} = E_k$. Thus $\beta_{kj} = 0$, and $\alpha_{kj} = E_k E'_{qj}$.

The corrected kinetic energy [10], which contributes to system separation, is given by

$$V_{KEcorr} = \frac{1}{2} M_{eq} \omega_{eq}^2 \quad (8)$$

and the transient energy becomes

$$V = V_{KEcorr} + V_{PE} \quad (9)$$

V. ESTIMATION OF E'_q AND E'_d AT THE PEAK POINT

When the generator flux, and hence its internal EMF, are not constant a great deal of complexity is added to the determination of the critical energy [2-7] as compared with the case of the classical model when the generator internal EMF is considered constant. In the latter case the potential energy surface is defined only in the angular space. The key issue is to determine the point on that surface which represents the controlling unstable equilibrium point (UEP) for the disturbance under investigation. The mode of disturbance test identifies the particular generators whose rotor angles, in the UEP, generally exceed $\pi/2$. The potential energy at that UEP represents the critical transient energy with which the energy at the end of the disturbance is to be compared for transient stability assessment [10,11].

The recently developed transient energy function [1] assumes a changing generator flux, even though the internal voltage E' is considered constant at its average value in the first swing. The changing value of E' is dealt with as a "parameter variation". Thus the potential energy surfaces are not totally determined in the angle space alone; rather their shape is affected by the value of E' . Furthermore, at the peak of the potential energy on the critically stable trajectory (referred to as the peak point in this paper), the derivatives of some of the state variables (e.g., $\dot{E}'_q, \dot{E}'_d, \dot{E}'_{FD}$) may not be zero. This represents a major difference between the case when the exciter effects are included and that when the classical model is used.

The critical value of the transient potential energy sought must therefore be a maximum not only with respect to the vector of generator rotor angles θ , but also with respect to the internal generator voltages E' for the generators represented in detail. The point of interest at which the potential energy is a maximum (the peak point) is not an equilibrium point, i.e., not a UEP, since some derivatives may not be zero. However, the UEP near it must represent a limiting condition.

Another complication to be considered is the situation when a number of generators are electrically close to the disturbance. If some of them are equipped with modern exciters the mode of disturbance, i.e., the generators whose rotor angles are greater than $\pi/2$ in the critical energy computation, may be different from the case of the classical model.

Approximate Function For $E'_q(\theta)$

Attention will first be focused on the voltage E'_q , which the stator voltage corresponding to the flux linking the main field winding. It is also this voltage which dominates the magnitude of the voltage E' behind transient reactance. The concept of time is

lost in the direct method. The only available information is the value of the parameters at fault clearing and at the limiting conditions. Let the value of E'_q at fault clearing E_q^{cl} and at the limiting condition E_q^{lim} be known. During the first swing, we assume that the rotor position θ and $E'_q(\theta)$ have the approximate exponential relation of

$$E'_q(\theta) = E_q^{lim} - (E_q^{lim} - E_q^{cl}) e^{-\frac{\theta - \theta^{cl}}{\Delta\theta}} \quad (10)$$

where $\Delta\theta$ is estimated from conditions at or near fault clearing

$$\Delta\theta = (E_q^{lim} - E_q^{cl}) / \left(\frac{dE'_q}{d\theta} \right)_{cl} \quad (11)$$

This approximation was assumed after observing several time simulation runs for the behavior of E'_q . Judicious choice of $\Delta\theta$ determines the point on the $E'_q(\theta)$ curve corresponding to conditions near the peak point. By experimenting with several faults on two test power networks it was found that a good estimate of E'_q near the peak is obtained when $\frac{dE'_q}{d\theta}$ in (11) is estimated at a time interval prior to fault clearing.

For the choice of E_q^{lim} , the value of E'_q at the UEP would be a logical choice, since the UEP solution represents conditions where the derivatives of the state variables are equal to zero. A simplification to the limiting (or UEP) conditions is to use the value of E'_q at the so-called "corner point," instead of that at the UEP, in (10). The corner point (CRN) is obtained by solving [12]:

$$\dot{E}'_{qi} = 0, \dot{E}'_{di} = 0, \dot{E}'_{FDi} = 0, \dot{\omega}_i = 0 \quad (i=1,2,\dots,m) \quad (12)$$

for generators equipped with exciters.

E'_q essentially represents the effect of armature reaction, and from several time simulation runs was found to have a fairly linear behavior between fault clearing and the peak point.

Peak Point Estimation by Using Approximate $E'_q(\theta)$

When the mode of disturbance (MOD) is determined for the detailed model, the approximate peak point of each candidate mode, called the ray point, is obtained by maximizing the potential energy on the ray from the post fault stable equilibrium point to the corner point. This point is obtained by solving the one-dimensional search problem given by maximizing the potential energy on the ray from the post fault stable equilibrium point to the corner point.

This point is obtained by solving the one-dimensional search problem given by

$$\frac{dV}{d\alpha} \approx \sum_{i=1}^n P_{acc,i} (E'_q(\theta^{ray}), E'_d^{ray}, \theta^{ray}) \cdot (\theta_i^{CRN} - \theta_i^{SEP}) = 0 \quad (13)$$

where $\theta^{ray} = \theta^{SEP} + \alpha(\theta^{CRN} - \theta^{SEP})$,

$E'_q(\theta)$ is given in (5),

$$E_d^{ray} \approx \frac{1}{2} (E_d^{SEP} + E_d^{CRN}) = \text{const.}$$

The use of the approximate peak point makes the technique more practical. It results in a considerable saving of computer time because the approximate peak point is determined for all the candidate modes.

Having determined the correct M.O.D. using the approximate peak point, the transient stability

assessment is made by obtaining the exact peak point. This is done by solving,

$$P_{acc,i}(E_q^-(\theta^P), E_d^P, \theta^P) = 0 \quad (i=1,2,\dots,n) \quad (14)$$

where

$$E_d^P \approx (E_d^{SEP} + E_d^{CRN})/2 = \text{const.}, \text{ and} \\ E_q^-(\theta) \text{ is given in (10).} \quad (15)$$

From the above, we note that the peak point plays a similar role in the TEF method with a detailed generator model, as that of the UEP point in the classical model TEF method. It is used to evaluate the critical value of the transient energy function.

It is to be emphasized that $\dot{E}_q^P$, $\dot{E}_d^P$, and $\dot{E}_{FD}$ may not be equal to zero at the peak of the critical stable system trajectory, this makes the estimation of E_q^P and E_d^P computationally burdensome.

IV. DETERMINATION OF THE M.O.D. WITH DETAILED GENERATOR MODEL

Let us assume that the condition at the fault clearing and the post fault SEP point have been obtained.

To identify the correct mode of disturbance, the following procedure is used:

1. The candidate modes of disturbance to be checked are identified. This can be done either by inspection (if warranted by the system configuration), or by applying the automatic mode of disturbance test [11] used with the classical power system model. In the latter case, a suitable number of the candidate modes is retained for checking.
2. For the first mode selected, the rotor angles at the corner point [11] are obtained as:

$$\theta_j^{CRN} = \left. \begin{aligned} &\theta_j^S + (\pi - 2\theta_{eq}^S) \frac{M_{T-k}}{M_T} \quad (j \in \text{critical machine group } k) \\ &\theta_j^S - (\pi - 2\theta_{eq}^S) \frac{M_k}{M_T} \quad (j \in \text{rest of the system } (T-k)) \end{aligned} \right\} \quad (16)$$

where

$$\theta_{eq}^S = \theta_k^S - \theta_{T-k}^S, \quad \theta_k^S = (\sum_{i \in k} M_i \theta_i^S) / M_k, \quad \theta_{T-k}^S = (\sum_{i \in T-k} M_i \theta_i^S) / M_{T-k}$$

$$M_k = \sum_{i \in k} M_i, \quad M_{T-k} = \sum_{i \in (T-k)} M_i, \quad M_T = M_k + M_{T-k}.$$

In order to obtain the voltages for the detailed generator models, E_q^S , E_d^S are used directly as starting values of the CRN solution.

3. Using the procedure given in [12], the classical model generator angles are held at θ^{CRN} , and the detailed generator model E_q^{CRN} and E_d^{CRN} , E_{FD}^{CRN} , and θ^{CRN} are obtained by solving equations (12).
4. The approximate peak point, i.e., the ray point, is searched for using (13) for the candidate mode of disturbance.
5. For generator i the average values of E_q^P and E_d^P (for use in the expression for the energy function) is given by

$$E_q^P = \frac{1}{2} [E_q^{CL} + E_q^{ray}(\theta^{ray})], \quad E_d^P = \frac{1}{2} [E_d^{CL} + E_d^{ray}(\theta^{ray})] \quad (17)$$

For the values of θ^{ray} , E_q^P , and E_d^P obtained, compute ΔV_{PE} using θ^{CL} at the lower limit and θ^{ray} at the upper limit in Equation (6).

6. $V_{K.E.corr}$ is computed at fault clearing, then $\Delta V_{PEN} = \Delta V_{PE} / V_{K.E.corr}$ is obtained.
7. Steps 2-6 are repeated for the other candidate modes. The correct mode is that having the lowest ΔV_{PEN} .

VII. TRANSIENT STABILITY ASSESSMENT

Critical Value of Transient Energy

The critical value of the transient energy used in the final transient stability assessment is the value of the potential energy given by (6) evaluated at the rotor angles of θ^P in (10). In Reference [1] it was shown that a value of E_q^{av} and E_d^{av} gives an accurate accounting of the energy in the first swing, where

$$E_q^{av} = \frac{1}{2} (E_q^{CL} + E_q^P), \quad E_d^{av} = \frac{1}{2} (E_d^{CL} + E_d^P) \quad (18)$$

E_q^P and E_d^P obtained from (14) and (15).

Procedure for Transient Stability Assessment

From the analysis presented above, a procedure for transient stability assessment with exciter effect included is suggested as follows:

- Step 1: Conditions at fault clearing are calculated and the post fault admittance matrix is formed for later use.
- Step 2: The post fault SEP point is solved for.
- Step 3: Procedure for determining the mode of disturbance (M.O.D.) with the generator detailed model is conducted, and the starting point for the peak point solution is generated. In the procedure the corner point (CRN) is evaluated to obtain an approximate exponential function of $E_q^-(\theta)$. (See Section V.)
- Step 4: The critical trajectory exact peak point is found by solving $P_{acc,i}(E_q^-(\theta), E_d^P, \theta) = 0 \quad (i=1,2,\dots,n)$ where $E_{q1}^-(\theta_1)$ is the approximate function derived in Step 3 and $E_{di}^P \approx (E_{di}^{CL} + E_{di}^{CRN})/2 = \text{const.}$ The ray point searched in Step 3 is used as starting values for the exact peak point solution.
- Step 5: Transient stability assessment is made using the normalized energy margin $\Delta V_{n|cl}^{\text{peak}}$.

VIII. VALIDATION STUDIES

In this section the results of the previous two sections, i.e., the conditions at the peak of the trajectory, the critical value of the transient energy, and their use in transient stability assessment are compared with data obtained with time simulation using the EPRI RP745 transient stability computer program (also known commercially as DYNSTAB).

Test Networks

Two test networks are used in the validation studies: a 20-generator, 50-bus equivalent derived from the B.C. Hydro Network, and a 17-generator, 162-bus equivalent known as the Reduced Iowa Network. These networks have been previously described [1,10]. Only some of the severely disturbed generators are equipped with exciters (as indicated in the data below). The rest of the generators are represented by the classical model. Unless otherwise indicated, the exciters used are modern, high ceiling, high response ratio exciters.

Conditions at the Peak of the Trajectory

For selected disturbances in the two test networks, the estimated values of the critical generator rotor angles as well as their internal voltages, using equations (14) and (15) are compared with the corresponding values at the peak of the critical trajectory obtained from time simulation using the transient stability computer program.

Table 1 gives the data for two disturbances on the 20-generator B.C. Hydro Network close to the generators equipped with exciters (#9,10); and for four disturbances on the 17-generator Iowa Network. We note that the estimated values of E_q^P at the peak point compare well with the actual values on the (hypothetical) critical trajectory. In addition, the estimated rotor angles correctly account for the advanced machines affected by the disturbance.

Table 1. Comparison of Estimated Values of E_q^P and θ^P With Values from Critically Stable Trajectory

Reduced Iowa System					
Case	Gen. with Exciter	Estimated Values		Values from Time Solution	
		E_q^P	θ^P	E_q^P	θ^P
Cooper	2	1.2667	163.46	1.3932	171.18
C.B. #3	10	1.2300	175.31	1.3126	116.08
	12	1.1428	153.77	1.3403	168.16
Raun	5	1.3477	166.43	1.2452	119.39
	6	1.0890	169.35	1.1590	165.71
Ft. Cal.	16	1.1561	129.61	1.2871	197.56
B.C. Hydro System					
GMS500	9	1.2371	133.05	1.2209	129.70
	10	1.3373	137.33	1.3646	116.71
WSN500	9	1.1926	128.82	1.1761	123.80
	10	1.2996	131.86	1.3267	134.98

Effect of Exciter on Mode of Disturbance

Results for the effect of the exciter on the Mode of Disturbance are shown in Tables 2 and 3 for the 17-generator network and the 20-generator B.C. Hydro Network respectively.

The results in Tables 2 and 3 show that the presence of the exciter on some of the generators usually affects the transient energy margin computations, and may change the mode of disturbance. For the Iowa Network, only in one case (fault at Cooper) does the mode of disturbance appear to be susceptible to change. When the classical generator model is used the mode is clearly obtained with only generator no. 2 advanced; while when the exciter effects are included either one of two modes, #2 or #2 and 17, would essentially give the same values of $\Delta V_{PE n}$.

The results for the B.C. Hydro Network show that the exciter effects are much more pronounced for that network. The mode with the lowest normalized potential energy changes. With the classical model, the mode is clearly generators #9 and 10. When the exciter is used, however, the mode becomes generators no. 3, 4, 9, and 10 advanced. Both modes result in good transient stability assessment, i.e., give a critical clearing time comparable to that obtained with time simulation.

The results in Tables 2 and 3 seem to show that the procedure proposed in this paper is adequate for dealing with the possible effect of the exciter on the mode of disturbance, i.e., checking only some modes with low values of $\Delta V_{PE n}$ more than 50% higher than the lowest need not be checked. This is also indicated in Table 2 in the C.B. #3 and Raun faults, where only one mode is listed in the Table.

Transient Stability Assessment Results

The suggested procedure for transient stability assessment is tested on both the reduced Iowa and B.C. Hydro systems.

On the Iowa system, four test cases are studied, i.e., three phase faults at Cooper, C.B. No. 3, Raun, and Ft. Calhoun respectively.

On the B.C. Hydro system two test cases are studied, i.e., three phase faults at GMS500 and WSN500 respectively.

Table 2. Effect of Exciter on M.O.D. 17-Gen. (Iowa) Network

CASE	t_{cl} (sec.)	TESTED MODE	$\Delta V_{PE n}$		$V_{KE corr}$	FINAL M.O.D.
			CLASSICAL MODEL	DETAILED MODEL		
Cooper (#2 with exciter)	0.15	2 2,17 2,17,10,12	2.8674 3.3473 4.5536	3.3757 3.4022 4.5288	3.1677 4.5963 5.2530	#2
C.B. #3 (#10,12 with exciter)	0.10	10,12 10,12,17 10,12,17 6,5,2,16	6.3475 10.2986 --- 13.7494	7.4022 --- --- ---	1.7074 1.9919 --- 2.1909	#10,12
Raun (#5,6 with exciter)	0.10	5,6 5,6,10,12 5,6,10,12,17	4.9070 10.3330 11.5022	5.6677 --- ---	3.0791 2.6073 2.2523	#5,6
Ft. Cal. (#16 with exciter)	0.15	2,5,6,10 12,16,17 2,5,6,8,10, 12,16,17 5,6,10,12 16,17	7.4297 --- 8.0093 8.7117	7.5392 --- 8.1310 8.9142	3.8062 3.7633 3.5508	#2,5,6,10 12,16,17

Table 3. Sample of M.O.D. Conversion with Exciter Effect Included (20-Gen., B.C. Hydro Network, Fault on WSN500)

CASE	t_{cl} (sec.)	TESTED MODE	$\Delta V_{PE n}$		$V_{KE corr}$	FINAL M.O.D.
			CLASSICAL MODEL	DETAILED MODEL		
M.O.D. Test	0.05	9,10 3,4,9,10	0.6220 1.0568	4.8472 4.6926	0.9886 0.9885	M.O.D.= 3,4,9,10
TEF Evaluation	0.100	9,10	---	1.2669	3.9246	$t_{cr}(TEF)=$ 0.113" (M.O.D.=9,10)
	0.110	9,10	---	1.0643	4.7233	
	0.100	3,4,9,10	---	1.2096	3.9139	$t_{cr}(TEF)=$ 0.111" (M.O.D.=3, 4,9,10)
	0.110	3,4,9,10	---	1.0201	4.7058	

*From time simulation: $t_{cr}(\text{stable/unstable}) = 0.107/0.110$ sec.

The parameters of the detailed generator model and exciter are listed in Table 4.

Results of transient stability assessment, and comparison with time simulation are shown in Table 5. Critical clearing times are used in this series of tests as a figure of merit for comparison purposes. These are calculated by interpolating the clearing times to obtain a $\Delta V_n = 0$.

Comparison With the Classical Generator Model

To put the results presented below in perspective, a word about the meaning of the classical generator model is in order.

The classical generator model assumes constant flux linkage, hence constant internal voltage, during the first swing. This model may give conservative or optimistic results, as compared to the detailed generator-exciter model, depending on the fault duration and the type of excitation system used (see [8], Chapter 8).

For a short fault duration (e.g., less than 0.1 s) the classical model corresponds to a relatively slow, low response ratio exciter. Thus, when the generators near the fault are equipped with fast, high response ratio exciters, the classical model gives rather conservative results, i.e., lower power limits or shorter critical clearing times.

The situation is reversed with long fault durations (e.g., $t_{cf} > 0.2$ s). Constant flux linkage with such long faults corresponds to a very modern (high ceiling, high response ratio) exciter. Unless such an exciter is actually in use, the classical model would actually give optimistic results. These points should be kept in mind when exciter effects are modeled, and are compared with the results obtained with the classical generator model.

The comparison of the critical clearing time obtained by using the detailed model with that obtained by using the classical model is included in Table 6.

Effect of Exciter Parameters

The effect of exciter gain and time constant was investigated on the WSN500 bus cleared by opening line WSN500-KLY500. The same fault was repeated with the exciter having a gain of 45, time constant $\tau_e = 0.5$ second, and a lower $E_{FD\text{limit}}$. The time solution showed

that the system became unstable. The results are shown in Table 7, where ΔV_n is the normalized energy margin.

Effect of Exciter on Power Limits

A three-phase fault was applied at WSN500 (B.C. Hydro Network), cleared at four cycles by opening line WSN500-KLY500. For this disturbance, the power generation level at the GMS station (#9) was found by repeating the stability analysis using the TEF method until critically stable and unstable conditions are reached.

The test was conducted with the GMS and P. Canyon generator equipped with high ceiling, high gain exciters (same parameters as in Table 4), and when the same two generators are modeled classically. Table 8 summarizes the results of this test. It is clear that, for this short duration fault, the modern exciter substantially increases the power limit.

IX. DISCUSSION AND CONCLUSIONS

A transient energy function has been successfully developed in [1] which can account for the exciter effects: both situations where the flux increases as well as conditions of flux decay. In this paper a procedure is suggested to determine its critical value. This procedure is based on: i) estimating the values of the internal generator voltages and angles at the peak point of the critical trajectory, and ii) determining the effect of the exciter on the modes of disturbance (M.O.D.) with the lowest normalized potential energy, determined by the conventional (classical) M.O.D. test. Preliminary validation studies on two test power networks indicate that the proposed procedure gives a reasonably good estimate of the conditions at the peak point of the critical trajectory, and the critical value of the transient energy.

The newly developed transient energy function and its critical value have been successfully used for transient stability assessment in the two test networks. The assessment included: correct stability classification, and reasonably accurate critical clearing times and transient stability power limits. In addition, the results obtained correctly reflect the exciter effects when "strong," modern exciters are used as compared with cases when the generators are equipped with "ordinary" exciters. Thus, the newly developed transient energy function accurately accounts for both "flux boosting" and "flux decay" conditions.

Table 4. Parameters for the Detailed Model Generators and its Exciters of the Test Cases
(Time Constants: sec. Other Variables: p.u.)

CASE	Gen. With Exciter	x_d'	x_q'	x_d	x_q	τ_{d0}'	τ_{q0}'	τ_e	K_e	$E_{FD\text{ max}}$	$E_{FD\text{ min}}$
Cooper	2	0.0437	0.0437	0.1807	0.1755	6.54	1.09	0.05	145	8.85	-8.85
C.B. #3	10	0.1659	0.1659	1.1375	1.0901	4.56	0.76	0.05	145	8.85	-8.85
	12	0.0297	0.0297	0.2207	0.2108	3.93	0.65	0.05	145	8.85	-8.85
Raun	5	0.0541	0.0541	0.3570	0.3409	4.43	0.74	0.05	145	8.85	-8.85
	6	0.0197	0.0197	0.1407	0.1354	4.00	0.67	0.05	145	8.85	-8.85
Ft. Cal.	16	0.0559	0.0559	0.3165	0.3013	7.60	1.27	0.05	145	8.85	-8.85
GMS500	9	0.0075	0.0075	0.0466	0.0222	4.90	0.06	0.05	145	8.85	-8.85
	10	0.0367	0.0367	0.1005	0.0611	4.73	1.74	0.05	135	7.50	-7.50
WSN500	9	0.0075	0.0075	0.0466	0.0222	4.90	0.06	0.05	145	8.85	-8.85
	10	0.0367	0.0367	0.1005	0.0611	4.73	1.74	0.05	135	7.50	-7.50

Table 5. Transient Stability Assessment Results on the Test Networks

FAULT BUS	TRIP LINE	GEN. WITH EXCITER	M.O.D.	t_{cl}	$V_{KE,corr}$	ΔV_n	CRITICAL CLEARING TIME		
							t_{cr} (TEF)	TIME SIMULATION ST./UNST.	
Cooper (#6)	6-439	#2	#2	0.200 0.205	5.6196 5.9019	0.0662 -0.0730	0.2024	0.219	0.222
C.B. #3 (#436)	436-771	#10,12	#10,12	0.180 0.185	5.3849 5.6801	0.1011 -0.0438	0.1835	0.194	0.195
Raun (#372)	372-193	#5,6	#5,6	0.165 0.170	8.0186 8.4881	0.1281 -0.0289	0.169	0.179	0.182
Ft. Cal. (#773)	773-779	#16	#2,5,6,10, 12,16,17	0.345 0.355	9.6418 9.7882	0.0718 -0.0641	0.350	0.348	0.351
GMS500	GMS500-WSN500	#9,10	#9,10	0.110 0.120	6.2531 7.4398	0.0707 -0.1467	0.113	0.125	0.130
WSN500	WSN500-KLY500	#9,10	#3,4,9,10	0.100 0.110	3.9139 4.7058	0.2096 0.0201	0.111	0.107	0.110

Table 6. Comparison of the Detailed Model TEF Method Results with the Classical Model TEF Method Results

FAULT BUS	TRIP LINE	GEN. WITH EXCITER	DETAILED MODEL			CLASSICAL MODEL		
			M.O.D.	t_{cr} (TEF)	t_{cr} (T.S.) st./unst.	M.O.D.	t_{cr} (TEF)	t_{cr} (T.S.) st./unst.
Cooper	6-439	#2	#2	0.202	0.219/0.222	#2	0.2050	0.216/0.220
C.B. #3	436-771	#10,12	#10,12	0.184	0.194/0.195	#10,12	0.187	0.1923/0.192
Raun	372-193	#5,6	#5,6	0.169	0.179/0.182	#5,6	0.197	0.200/0.204
Ft. Cal.	773-779	#16	#2,5,6, 10,12, 16,17	0.350	0.348/0.351	#2,5,6, 10,12, 16,17	0.350	0.345/0.356
GMS500	GMS500-WSN500	#9,10	#9,10	0.113	0.125/0.130	#9,10	0.065	0.070/0.074
WSN500	WSN500-KLY500	#9,10	#3,4, 9,10	0.111	0.107/0.110	#9,10	---	< 2 cycles

Table 7. B.C. Hydro Network-Effect of Exciter Parameters on Transient Stability

Fault Location	WSN500, Line WSN500-KLY500 Cleared	
Exciter Data		
$K_A(9,10)$	145,135	45,45
τ_E	0.05	0.50
$E_{FDlimit}(9,10)$	8.85, 7.5	4.0,4.0
Time Solution:		
t_{cr} (st./unst.)	6.4~/6.6~	5.6~/5.8~
Mode of Dist. (Adv. Gens.)	(9,10)	(9,10)
$t_{cr}^{\sim}$	6.4~	5.6~
TEF ΔV_n	0.0904	-0.2473

Table 8. Power Limits Test (B.C. Hydro System)

Fault: WSN500, $t_{cl}=4\sim$

Line Cleared: WSN500-KLY500

System Condition	Power at GMS	
	With Exciter	Classical
Stable	2374 MW	2030 MW
Unstable	2384 MW	2040 MW

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REFERENCES

- [1] A. A. Fouad, et al., "Extending Application of the Transient Energy Function Method," EPRI Report EL-5215, RP2206-5, by the Electric Power Research Institute, September 1987.
- [2] T. Athay, et al., "Transient Energy Analysis," System Engineering for Power, Emergency Operating State Control, Section IV, U.S. Dept. of Energy, Publication No. CONF-790904-PI.
- [3] H. Sasaki, "An Approximate Incorporation of Field Flux Decay Into Transient Stability Analyses of Multimachine Power Systems by the Second Method of Lyapunov," IEEE Trans. on PAS, Vol. PAS-98, No. 2, March/April 1979, pp. 473-483.
- [4] N. Kakimoto, Y. Ohsawa, and M. Hayashi, "Transient Stability Analysis of Multimachine Power Systems With Field Flux Decays via Lyapunov's Direct Method," IEEE Trans. on PAS, Vol. PAS-99, No. 5, September/October 1980, pp. 1819-1827.
- [5] N. A. Tsolas, A. Arapostathis, and P. P. Varaiya, "A Structure Preserving Energy Function for Power System Transient Stability Analysis," IEEE Trans. on CAS, Vol. CAS-32, No. 10, October 1985, pp. 1041-1049.
- [6] Sastry, H. S. Y., "Application of Topological Energy Functions for the Direct Stability Evaluation of Power Systems, a Ph.D. thesis submitted to the Department of Electrical Engineering, Indian Institute of Technology, Kanpur, May 1984.
- [7] Bergen, A. R. and Miyage, H., "Stability Studies of Multimachine Power System with Effects of Automatic Voltage Regulators," IEEE Trans. on Automatic Control, March 1986, pp. 210-215.
- [8] P. M. Anderson and A. A. Fouad, Power System Control and Stability, Iowa State University Press, 1977.
- [9] R. G. Brown, Introduction to Random Signal Analysis and Kalman Filtering, John Wiley & Sons, 1983.
- [10] A. A. Fouad, et al., "Transient Stability Margin as a Tool for Dynamic Security Assessment," Electrical Power Research Institute Report EL-1755, March 1981.
- [11] M. A. El-Kady, A. A. Fouad, et al., "Direct Analysis of Transient Stability for Large Power System," EPRI Report EL-4980, RP2206-1, by the Electric Power Research Institute, December 1986.
- [12] Y-X. Ni, C. G. Shin, and A. A. Fouad, "TEF Method Solution with Exciter: Numerical Technique Used," to appear in the Proceedings of the 1987 North American Power Symposium, held in Edmonton, Alberta, October 1987.

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